

SciForge: A Multimodal Research-Native AI Workbench for Scientific Discovery

Layer 1

Desktop Agent

Local execution
Workspace context

Group Chat Room

Shared room
Lead-agent structure

Mobile Remote

Remote supervision
Approval control

Multi-Channel Request

Layer 2

Research Applications

Literature Inbox

Curate papers into project knowledge

Artifact Review Canvas

Review artifacts through visual feedback

Presentation Studio

Shape research into polished talks

Research Ideation & Decision

Steer hypotheses and research choices

Manuscript Workbench

Write with evidence-linked claims

Research Task

Layer 3

Agent Runtime

Coordinates agents and tools for long-running research tasks

Goal-Centered Memory

Preserves goals, decisions, and project state across sessions

Core Task Engine

Automation Engine

Turns research protocols into reusable automated pipelines

Governance Engine

Verifies evidence, provenance, and approval gates

Task Formation

Layer 4

Task Input Info

Multi-Scale Scientific Data Modalities
(Text, Image, Files etc.)

Dispatch

ESM: Sequence Expert

For Protein Sequences (FASTA)

Prot2Text: Structure Expert

For Protein 3D structures (PDB/mmCIF)

BioT5: Molecule Expert

For Small Molecules (SMILES/SDF/MOL)

C2S: Single Cell Expert

For Single-Cell Expression (H5AD)

Collect

Task Output Info

Provenance-rich, Modality-normalized Evidence Objects

Memory Cache

Memory Cache

Scientific Multimodal Model Router

Support

Layer 5

Local Scientific Memory

Persist artifacts, indexes, traces, and project state

Modular Capability System

Packages, SKILLS, MCP servers, plugins, and reusable tools

Infrastructure

Scientific Connectors

Bridge files, APIs, databases, instruments, and software

Team Workspace Sync

Synchronizes shared context, tasks reviews and supervision

Local-First Architecture Design

Privacy-preserving, all data stay on user's machine.